Analysis of SupplyChain Management

1. Executive Summary

This report presents a comprehensive analysis of delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. Out of 172,765 orders analyzed, 54.71% were delivered late, resulting in approximately $2.1M in profit at risk. While total profit reached $7.5M, persistent delays significantly impact both profitability and customer experience. A Random Forest model was developed to predict late deliveries, achieving ~74% accuracy in identifying high-risk orders before shipment.

2. Business Problem

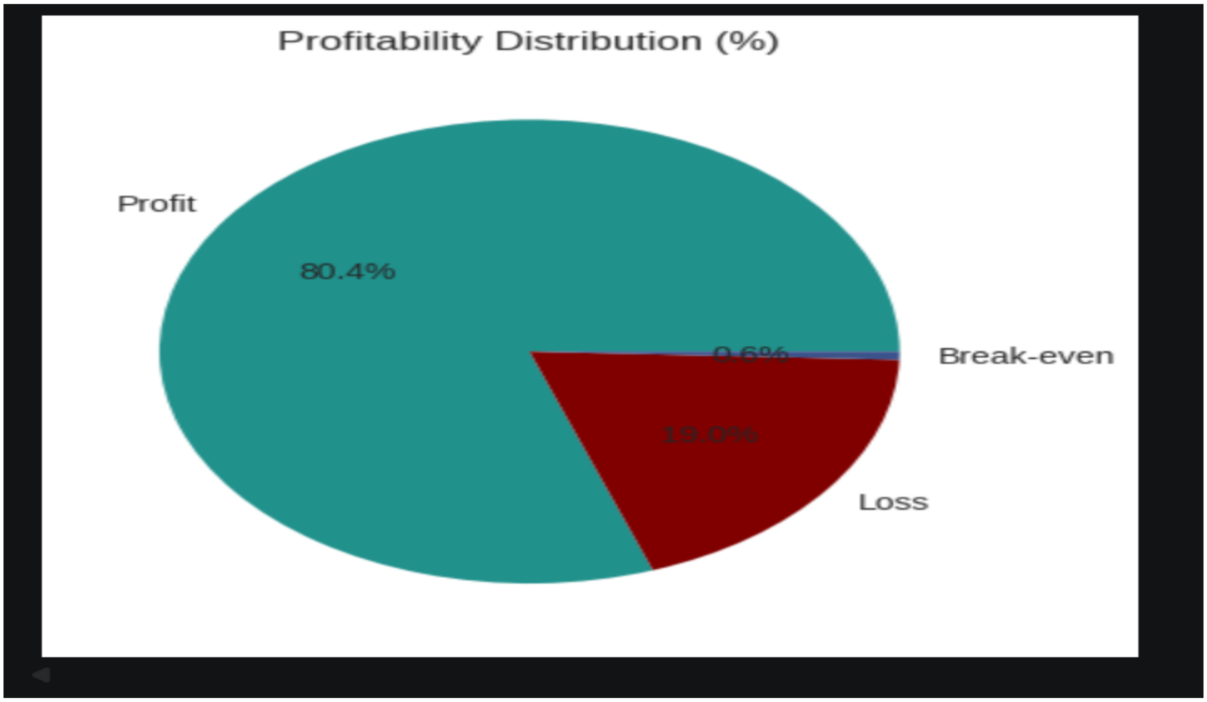
Actual shipping times frequently deviate from scheduled delivery windows, leading to reduced customer trust, lower profitability, and operational inefficiencies.

3. Key Performance Indicators

- Total Orders: 172,765 - Late Delivery Rate: 54.71% - On-Time Delivery Rate: 45.29% - Total Profit: $7.5M - Profit at Risk: $2.1M - Avg Delay (90th percentile): 3 days - Model Accuracy: ~74%

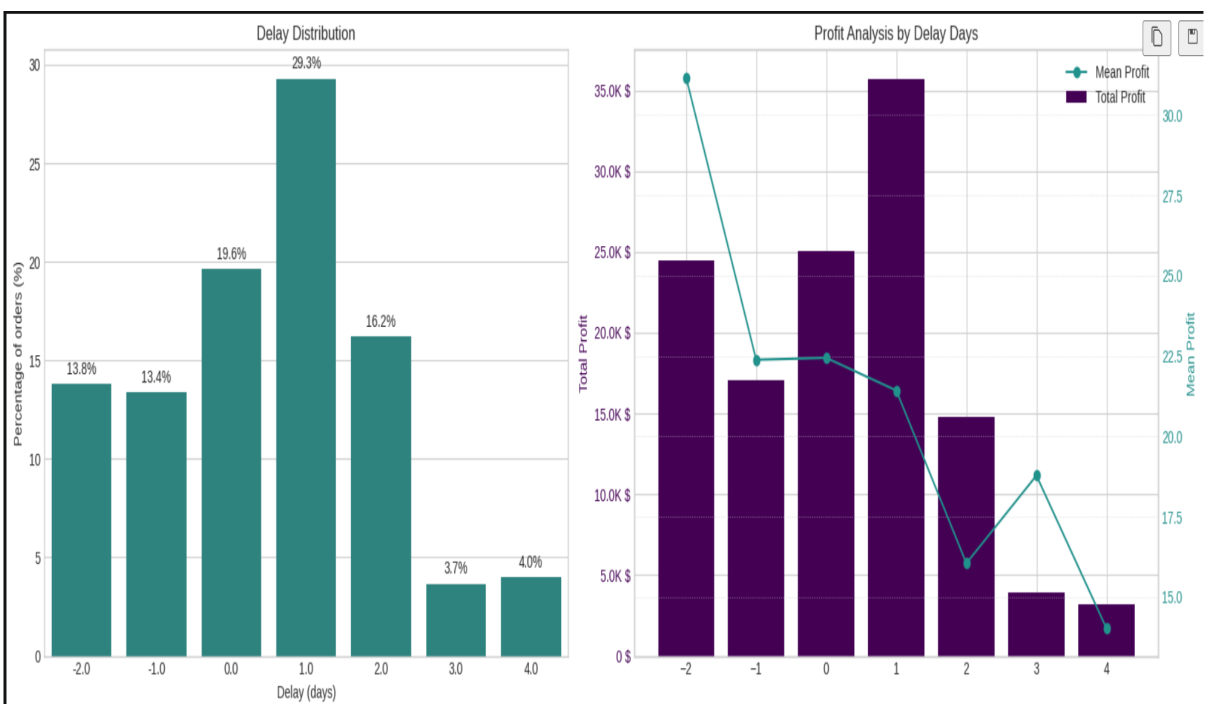
4. Profitability Analysis

80.4% of orders are profitable, while 19% result in losses. Loss-making orders are heavily concentrated among delayed shipments, indicating a strong correlation between delay and profitability erosion.



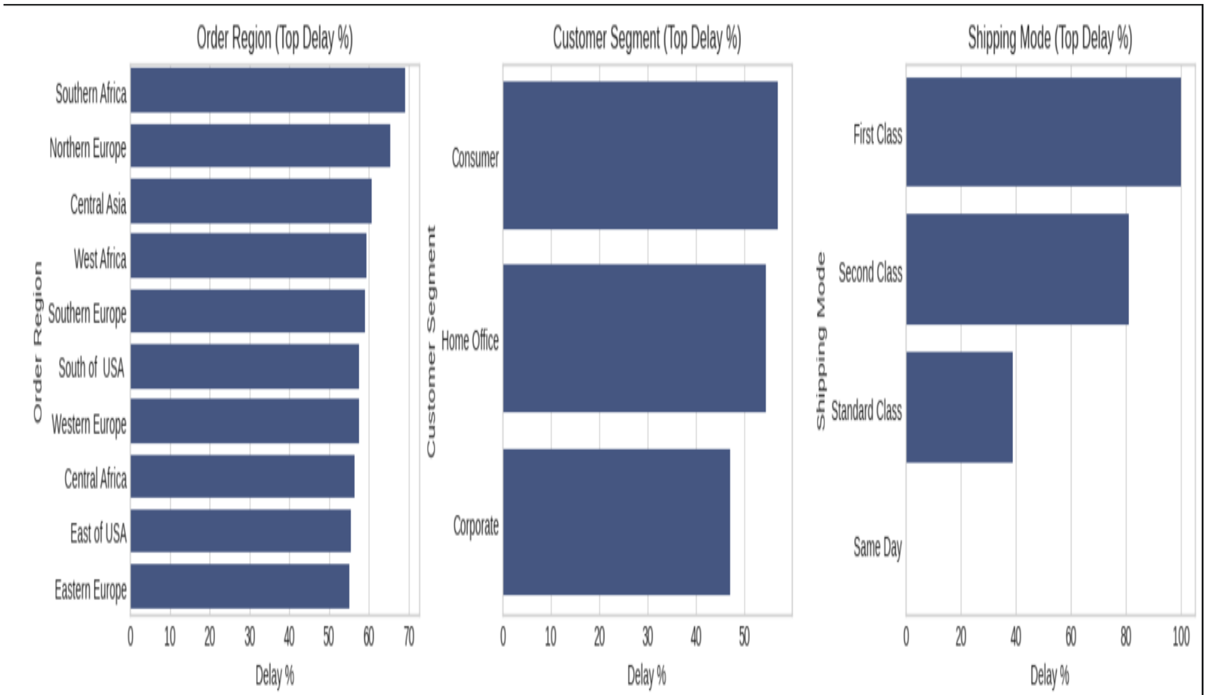
5. Delay Distribution & Impact

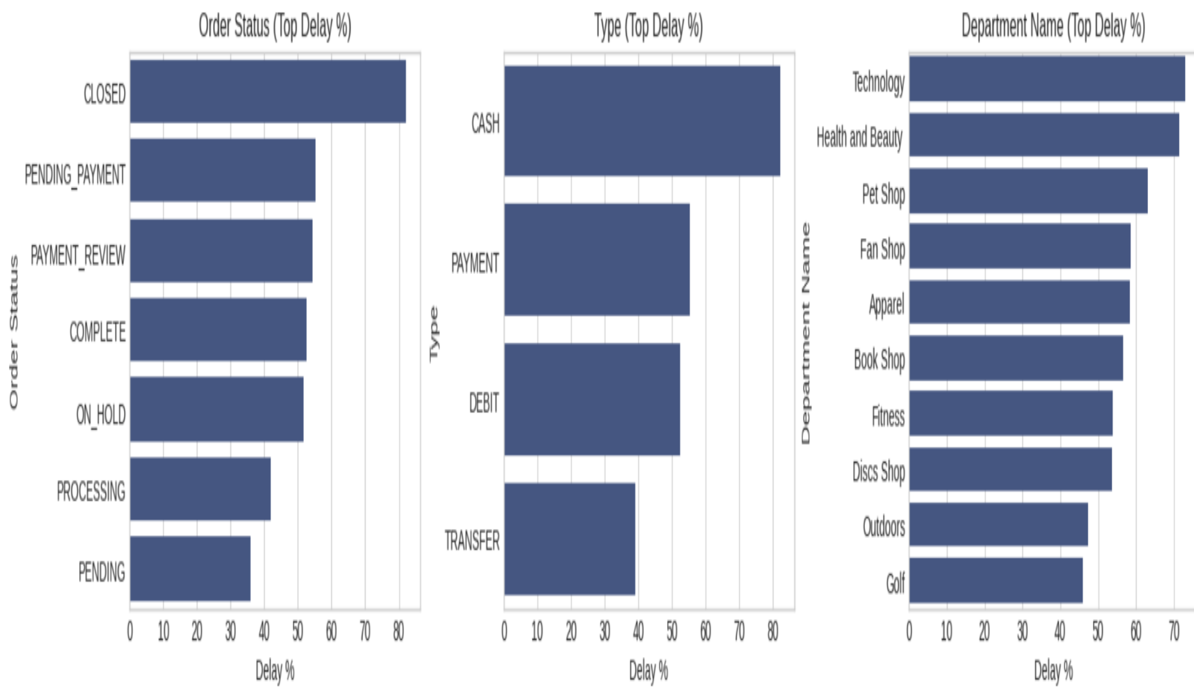
Approximately 29% of orders are delayed by 1 day, forming the largest segment. Orders delayed between 1–4 days account for the majority of delays. Profit decreases as delay increases, showing a clear negative relationship between delay duration and business value.



6. Bottleneck Detection

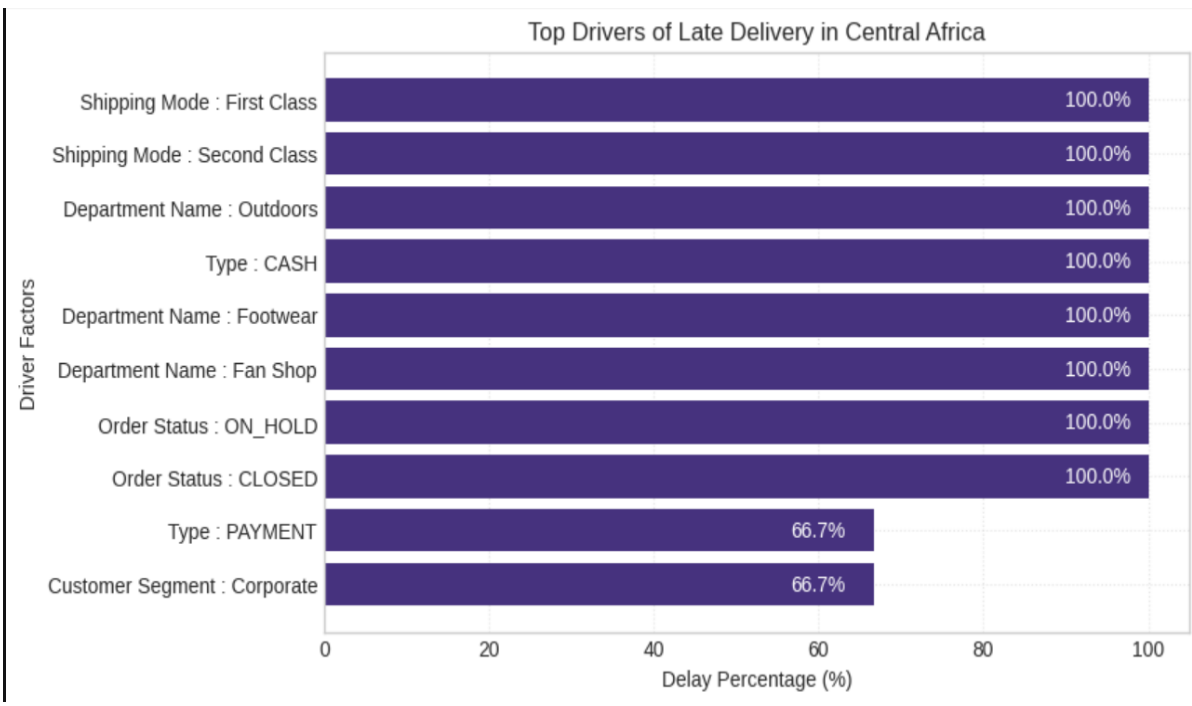
Shipping mode is the biggest contributor to delays: - First Class: ~100% delay rate - Second Class: ~80% - Standard Class: ~40% Regional differences are minimal (55–59%), indicating a systemic issue rather than localized problems.





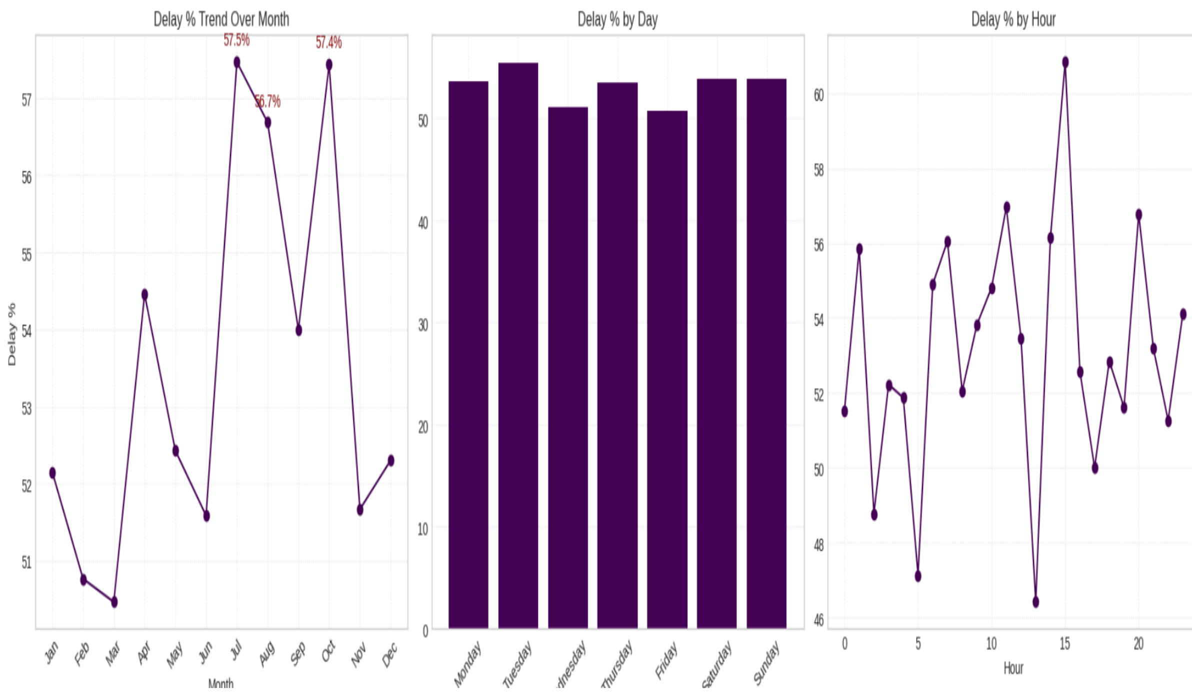
7. Root Cause Analysis

Central Africa shows the highest delays (~58.7%). Key drivers include: - Shipping Mode inefficiencies - Payment delays (Pending/Review) - Specific product categories (Outdoors, Golf) These indicate process-level failures rather than isolated issues.



8. Time-based Trends

Delay rates peak in July and October (~57%). Weekly patterns are consistent, while hourly trends show spikes around midday, suggesting operational bottlenecks during peak hours.



9. Predictive Model

A Random Forest classifier was trained to predict late deliveries. The model achieved: - Accuracy: ~66–74% - Balanced precision and recall This enables proactive identification of high-risk orders before dispatch.

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Classification Report:

              precision    recall  f1-score   support

           0       0.61      0.71      0.66       517
           1       0.72      0.62      0.67       622

    accuracy                           0.66      1139
   macro avg       0.67      0.67      0.66      1139
weighted avg       0.67      0.66      0.66      1139
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10. Strategic Recommendations

1. Audit First & Second Class shipping immediately 2. Deploy predictive alert system in production 3. Fix payment processing delays 4. Prepare seasonal surge capacity plans 5. Default eligible orders to Standard Class 6. Audit high-delay departments 7. Reduce loss-making orders 8. Continuously retrain predictive model

11. Conclusion

The analysis reveals a systemic delivery issue affecting over half of all orders. With $2.1M at risk, immediate operational intervention is required. Shipping mode misconfiguration is the dominant root cause, followed by payment bottlenecks. The predictive model provides a scalable solution to proactively reduce delays and improve profitability.